

Key Measurements and Interlocks

Experimental signals used for control, monitoring, and later analysis. The table keeps hardware context without listing implementation-level field names.

Quantity	Sensor or interface	Use in the experiment
TMI	Thermocouple at the HFE control point	Primary liquid-temperature feedback for the HFE cooling goal and LN valve auto mode.
THI	Heat-exchanger inlet thermocouple	Cold-side guard temperature used to prevent excessive heat-exchanger cooling.
TTI / TTO	Tank inlet and outlet thermocouples	Temperature difference across the tank loop and general cooldown monitoring.
PMI / PMO	Pressure transducers before and after the pump	Pump pressure rise; delta-P above 8.0 bar trips the firmware emergency stop.
PTA	Tank pressure transducer	Tank absolute pressure monitoring during filling, cooldown, and circulation.
Flow meter	MFC400 serial flow meter	Independent flow, density, and fluid-temperature reference during circulation.
VFD feedback	Pump drive serial monitor	Drive frequency, current, voltage, power, and estimated speed for pump-performance checks.
Scale	Independent USB scale reader	Liquid-nitrogen mass trend attached to telemetry for cooldown accounting.
Actuators	Pump PWM, LN valve relay, heater relays	Commands applied by the firmware after manual requests and safety checks.